10BASE-T1S, Physical Layer Collision Avoidance, OPEN Alliance TC14, and TC10 sleep/wake

1. Real problem this technology is solving

At the edge of a vehicle, many devices do not need one hundred megabits per second or one gigabit per second.

A door module, a seat controller, a lighting controller, a simple sensor cluster, or a local actuator island often needs modest bandwidth, low wiring cost, and simple integration.

Historically, that is where Controller Area Network (CAN) and Local Interconnect Network (LIN) lived.

The attraction of 10BASE-T1S is that it brings Ethernet framing and Ethernet tooling down into that low-cost edge domain without forcing every low-end node to use a switched point-to-point 100BASE-T1 or 1000BASE-T1 link.

Public IEEE and OPEN Alliance material describe 10BASE-T1S as a 10 megabit per second single-pair Ethernet physical layer that can operate point-to-point or as a shared multidrop network, which is exactly why it is interesting for zonal edge aggregation.

The key idea is this: 10BASE-T1S is not “cheap Ethernet” in the casual sense.

It is Ethernet redesigned for a very specific physical situation, namely a short-reach single twisted-pair segment where multiple nodes may share the same medium.

That shared-medium part is the reason one cannot understand 10BASE-T1S properly unless one first understand what happens on a common bus when more than one node wants to talk.

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2. What 10BASE-T1S actually is

Clause 147 defines the 10BASE-T1S Physical Coding Sublayer (PCS) and Physical Medium Attachment (PMA) sublayer, which together make up the 10BASE-T1S Physical Layer (PHY).

Public IEEE draft text describes the PHY as supporting full-duplex or half-duplex point-to-point communication and also half-duplex multidrop communication over a single twisted pair, at an effective rate of 10 megabits per second.

The same public text describes multidrop operation with at least eight inline PHYs and short stubs, while OPEN Alliance automotive implementation material treats the reference automotive multidrop case as a 25 meter inline twisted-pair topology.

So there are really two different worlds hiding under one name:

1. **Point-to-point 10BASE-T1S**, where two nodes are connected directly.
2. **Multidrop 10BASE-T1S**, where several PHYs share one physical segment.

That second case is where the interesting engineering starts, because a shared medium immediately raises one question: if two nodes want to transmit at nearly the same time, who gets the cable first?

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3. Why shared medium is hard

A shared cable is not a switch.

A switch gives every link its own collision-free point-to-point channel.

A multidrop bus does not.

If two nodes launch energy onto the same cable at the wrong time, the waveforms interfere. In old Ethernet language, that is a collision.

That is why plain half-duplex shared Ethernet historically relied on Carrier Sense Multiple Access with Collision Detection (CSMA/CD): listen first, transmit if the line is idle, detect collision if another station overlaps, stop, back off, and try again later.

This works, but it has a serious weakness for automotive networks: access time is no longer nicely bounded by a simple fixed schedule.

If several nodes become active together, collisions and random backoff can stretch latency and create unfairness.

Public IEEE Physical Layer Collision Avoidance material explicitly says this is the reason Clause 148 exists: to improve throughput, latency, and fairness on multidrop mixing segments while still sitting on top of the normal Ethernet medium-access-control world rather than replacing it with a totally different protocol.

That is the first deep conceptual point:

10BASE-T1S by itself gives you a shared Ethernet physical layer.
Physical Layer Collision Avoidance gives you a disciplined way to use that shared medium.

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4. Clause 147 fundamentals: what happens on the wire

Public IEEE Clause 147 text says 10BASE-T1S uses two-level Differential Manchester Encoding (DME) modulation at 12.5 megabaud, and it also uses Four-Bit/Five-Bit (4B/5B) coding in the Physical Coding Sublayer to improve Electromagnetic Compatibility (EMC) performance and support out-of-band signalling among connected PHYs.

In simpler language, the line is not carrying raw unsupervised logic bits. The PHY first shapes the information into a code that is easier to recover robustly and easier to push over a noisy automotive cable.

The line code in public 10BASE-T1S material is deliberately built for a short single-pair shared medium.

Differential Manchester Encoding is self-clocking, meaning timing information is embedded in the transitions themselves, and the public IEEE text explicitly notes that it is intrinsically balanced and helps robust clock and data recovery in noisy environments.

That matters in a car, where the electrical environment is much harsher than a quiet office patch panel.

Another important point is that Clause 147 already mentions optional support for Physical Layer Collision Avoidance, but that does not mean Clause 147 alone gives you collision-free multidrop behaviour.

Clause 147 is the physical layer foundation.

Clause 148 is the arbitration discipline layered above it for the half-duplex multidrop case.

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5. Clause 148 fundamentals: what Physical Layer Collision Avoidance really does

Clause 148 specifies an optional Reconciliation Sublayer for Physical Layer Collision Avoidance on top of Clause 147 PHYs operating in half-duplex multidrop mode.

Public IEEE material is very clear on three things.

1. Physical Layer Collision Avoidance is optional.
2. meant to improve throughput, fairness, and latency.
3. **not** a replacement for Carrier Sense Multiple Access with Collision Detection
4. **not** a replacement for Time-Sensitive Networking.

It works with the normal Ethernet machinery rather than replacing Ethernet with a brand new bus protocol.

The easiest way to understand Physical Layer Collision Avoidance is to think in terms of **permission to start transmitting**, not in terms of changing the frame format.

The frames are still Ethernet frames.

What changes is when the local Media Access Control is allowed to release a frame onto the shared medium.

Public IEEE material describes a cycle started by a **BEACON** from node identifier zero, followed by a sequence of per-node **Transmit Opportunities**.

Each node is statically assigned a unique identifier.

A node may begin transmission only when the current transmit opportunity matches its own identifier.

If a node has nothing to send, that opportunity quickly expires and the system advances to the next node.

That one design choice changes the network behaviour dramatically.

In plain half-duplex shared Ethernet, all nodes may try at once and collisions are resolved after the fact. In Physical Layer Collision Avoidance, the shared medium is pre-disciplined so only one node has the right to start at a given moment.

That is why public IEEE material can say there are no physical collisions on the line when Physical Layer Collision Avoidance is operating correctly.

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6. Why Physical Layer Collision Avoidance is not just Time Division Multiple Access

Time Division Multiple Access uses fixed slots whether anyone has data or not. Public IEEE Physical Layer Collision Avoidance material explicitly distinguishes the two.

A Physical Layer Collision Avoidance node keeps track of a short **Transmit Opportunity timer**, and if the node owning the current opportunity has nothing to send, the wasted time is just that short timer, not a full packet-sized slot.

That is why Physical Layer Collision Avoidance can preserve fairness and bounded access delay without wasting as much bandwidth as a classical fixed-slot scheme on a lightly loaded bus.

So the conceptual difference is:

- **Time Division Multiple Access** reserves full slots in advance.
- **Physical Layer Collision Avoidance** reserves only the right to start, and unused turns are yielded very quickly.

That difference is fundamental to why 10BASE-T1S is attractive for automotive edge networks. Many edge networks are bursty.

Sensors, local control traffic, diagnostics, and wake traffic are not all continuously active.

A rigid slotting system would waste bandwidth unnecessarily.

Physical Layer Collision Avoidance gives a cheaper compromise between “totally random access” and “hard scheduled bus.”

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7. The Physical Layer Collision Avoidance cycle

A Physical Layer Collision Avoidance cycle starts when the coordinator, which is always the node with identifier zero, sends a **BEACON**. That tells all other nodes, “reset your local transmit-opportunity counters; a new round begins now.”

The cycle then moves through node 0, node 1, node 2, and so on up to node N. Public IEEE material states that a Physical Layer Collision Avoidance cycle consists of one beacon and N+1 transmit opportunities, and that maximum latency is guaranteed to be less than one complete cycle.

Suppose node 3 has data ready. It waits as identifiers 0, 1, and 2 pass by. When transmit opportunity 3 arrives, node 3 may begin transmission.

If the Media Access Control side needs a little time before actual data appears on the line, the PHY may first send a **COMMIT** symbol sequence to hold the turn and prevent another node from assuming the opportunity was idle.

Public IEEE material explicitly describes **COMMIT** as the mechanism used to compensate for Media Access Control latency such as the interpacket gap.

If node 3 has nothing to send, it simply yields.

The opportunity timer expires, and the cycle moves on.

This is why the system stays efficient when only a few nodes are active.

The bus is not forced to wait through a full packet-sized reservation for an idle node.

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8. The configuration parameters that actually matter

The OPEN Alliance Physical Layer Collision Avoidance Management Registers specification gives the configuration knobs that engineers actually set in a real device. In the current public revision, the most important parameters are **NCNT**, **ID**, **TOT**, **MAXBC**, and **BTMR**.

Node identifier

The **ID** field is the local Physical Layer Collision Avoidance node identifier. Public OPEN Alliance text says it must be unique across the multidrop network to achieve collision-free operation. The special value 0 configures the node as the coordinator. The special value 255 suspends Physical Layer Collision Avoidance functions. That immediately tells you two failure modes to worry about during bring-up: duplicated node identifiers and nodes accidentally left in a suspended state.

Node count

The **NCNT** field sets the maximum number of nodes supported on the multidrop network.

On the coordinator, public OPEN Alliance text says it must be set at least to the number of nodes that may be plugged into the network.

It determines how many transmit opportunities exist in the cycle, and therefore it directly affects worst-case access latency.

If you set it too low, some nodes simply will not get a turn.

Transmit Opportunity timer

The **TOT** field sets the transmit-opportunity timer in bit-times.

Public OPEN Alliance material says this timer determines the opening of the transmit-opportunity window, must be set equally across all nodes for collision-free operation, and defaults to 32.

The same public text says the default guarantees proper operation for a network of conformant Physical Layer Collision Avoidance nodes over a 25 meter twisted-pair inline topology.

This is one of the most important engineering constraints on a 10BASE-T1S bus: identical timer assumptions across the segment are not optional.

Burst mode

The **MAXBC** field sets how many additional packets a node may send during its own transmit opportunity.

If it is set to zero, burst mode is off. The **BTMR** field sets how long the node waits for the Media Access Control to deliver another packet before yielding the current opportunity.

Public OPEN Alliance material says **BTMR** must be greater than the local Media Access

Control interpacket-gap duration plus any added transmit latency, and that the default value is sufficient for typical Media Access Control implementations. This is the tradeoff knob for asymmetric traffic: more burst capability improves efficiency for a busy producer, but it also lengthens the time other nodes may wait.

Diagnostics

The newer public Physical Layer Collision Avoidance register document also defines diagnostic bits. **RXINTO** indicates that a node received a packet during its own assigned transmit opportunity, which points to a duplicate node identifier on the segment. **UNEXPB** indicates an unexpected beacon on the coordinator, which points to more than one coordinator. **BCNBFTO** indicates a beacon arrived before the local node received its turn, which points to an **NCNT** misconfiguration on the coordinator. Those are exactly the kinds of failure signatures a real bring-up team needs.

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9. “OPEN Alliance TC14 version 1.5 automotive profile”

Public OPEN Alliance material does **not** present one single public specification literally titled “OPEN Alliance TC14 automotive profile version 1.5.” Instead, OPEN Alliance Technical Committee 14 publishes a **set** of 10BASE-T1S automotive-enabling specifications. The public committee page says the goal of Technical Committee 14 is to enable 10BASE-T1S in an automotive context through conformance, interoperability, and Electromagnetic Compatibility test specifications; advanced diagnostic features; a system implementation specification; a Physical Medium Dependent transceiver interface; Physical Layer Collision Avoidance registers; topology discovery; and collaboration with other committees on topics such as wakeup/sleep.

“The Technical Committee 14 automotive profile,” usually means the bundle of public and member specifications taken together. Publicly visible version numbers show that, for example, the **10BASE-T1S PMD Transceiver Interface** is version 1.5, and the **10BASE-T1S Half-Duplex Interoperability Test Suite** is also version 1.5, while the **10BASE-T1S System Implementation Specification** is version 1.0.

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10. Why OPEN Alliance Technical Committee 14 had to add more than the IEEE clauses

The IEEE clauses tell you how the Ethernet PHY and Physical Layer Collision Avoidance work. They do **not** fully tell you how to build an automotive controller plus transceiver combination that can be debugged, sleep, wake, support topology discovery, survive host resets safely, and pass interoperability tests across vendors. That is where OPEN Alliance Technical Committee 14 adds real value. The public committee goals, the public PMD

interface specification, and the public test-suite references all show that the automotive problem is broader than raw PHY conformance.

A good example is the public **10BASE-T1S Physical Medium Dependent (PMD) Transceiver Interface version 1.5** document. Its revision history explicitly says version 1.5 added topology discovery, receive-pin Return-to-Zero Inverted encoding, Energy Detect pin filtering, and wake/sleep support. That tells you what the industry found missing from the basic PHY clauses when trying to productize 10BASE-T1S in real vehicles.

The same public PMD interface document also explains how the host talks to the transceiver. In data mode, two pins convey received information to the host: the receive pin and the energy-detect pin. In configuration mode, those same pins become the Management Data Clock and Management Data Input/Output pins for management access. The document further says the Energy Detect pin reports collision-detection information during transmit and acts as a filtered window-comparator output during receive. That is not abstract theory. That is exactly the kind of low-level behaviour a driver and hardware team need in order to make a host controller and transceiver interoperable.

So the right way to think about OPEN Alliance Technical Committee 14 is this:

- IEEE gives you the shared-medium Ethernet foundation.
- OPEN Alliance Technical Committee 14 turns that foundation into an automotive integration ecosystem.

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11. TC10 sleep/wake: why automotive Ethernet needed its own power-state protocol

Classical Ethernet standards define how links communicate.

They do not, by themselves, define an automotive-style controlled low-power link shutdown and a fast, reliable, network-wide wake-up mechanism suitable for partial networking.

The public OPEN Alliance Technical Committee 10 sleep/wake specification states its objectives explicitly: comply with Carrier Sense Multiple Access with Collision Detection Media Access Control behaviour, comply with the media-independent-interface family, support global network wake-up including link start-up within less than 250 milliseconds, keep wake-up fully at physical layer level, support controlled selective shutdown, comply with AUTomotive Open System ARchitecture network management, and avoid unwanted wake-ups due to noise.

Sleep/wake is not just “power down the PHY.”
In an automotive network it must be a coordinated signalling system that avoids false wake-ups, supports partial networking, and works across PHYs and switches.

The classic public OPEN Alliance Technical Committee 10 specification defines three commands:

- **Low Power Sleep (LPS)**, which requests sleep
- **Wake-Up Request (WUR)**, which propagates a wake request over an active link
- **Wake-Up Pulse (WUP)**, which wakes a passive or sleeping link

The same public document explains the basic handshake: when one link partner requests sleep, it sends **Low Power Sleep**; the peer acknowledges with its own **Low Power Sleep**; both sides then move through silent states into sleep if the handshake completes correctly. If the handshake fails, the PHY returns to normal operation instead of leaving the link half-dead.

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12. Why 10BASE-T1S sleep/wake cannot simply copy point-to-point TC10

A 10BASE-T1S segment is not a simple point-to-point link.

It is a shared segment that may be quiet, partially active, or already carrying Physical Layer Collision Avoidance beacons.

Because of that, the 10BASE-T1S sleep/wake problem is different from the 100BASE-T1 or 1000BASE-T1 point-to-point case.

The public **10BASE-T1S Sleep/Wake-up Specification** says exactly this in substance:

IEEE 802.3cg does not define mechanisms for controlled 10BASE-T1S link-segment shutdown and wake-up, so the OPEN Alliance document adds a new Power Management client and related service primitives as a supplement to the 10BASE-T1S specification.

This public 10BASE-T1S sleep/wake document also changes the command model.

Instead of reusing the full classic **Low Power Sleep / Wake-Up Request / Wake-Up Pulse** set directly on the shared segment, it defines one segment wake-up command: **Wake-Up Pulse**.

The document also defines service primitives such as **LowPowerEntryLocal.request**, **WakeupLocal.request**, **Wakeup.request**, **Wakeup.indication**, **WakeupForward.indication**, and **WakeupForward.request**.

In the 10BASE-T1S sleep/wake model, these names are **service primitives** between the **Power Management Client** and the **Reconciliation Sublayer / Physical Layer**.

They exist because 10BASE-T1S needed an added mechanism for controlled low-power entry and wake-up on a shared segment, and the spec explicitly introduces this primitive set for that purpose.

The easiest way to understand them is to split them into three groups:

1. Local power-state control

- **LowPowerEntryLocal.request**
  This means: **"put my local PHY into low power."** The spec says its purpose is to shut down the Physical Layer in a controlled way **without corrupting ongoing transmissions on the link segment**. So this is not "turn the whole network off"; it is a request concerning the **local port/PHY**.
- **WakeupLocal.request**
  This means: **"wake my local PHY back up from low power."** It is a purely local wake request. Think of it as a wake caused by something local such as the Electronic Control Unit logic, ignition logic, or a local wake pin. Its purpose is only to transition the local PHY out of low power.

2. Segment-wide wake-up

- **Wakeup.request**
  This means: **"send a wake-up onto the 10BASE-T1S segment so that all supported nodes on that segment can wake."** The spec says this primitive requests that a **Wake-Up Pulse** be communicated to all nodes on the segment. It also says that if the device is itself in low power, this primitive implicitly means **WakeupLocal.request first, then Wakeup.request**. So this is the primitive for a **network-segment wake**, not just a local wake.
- **Wakeup.indication**
  This means: **"a wake-up event has been detected."** The spec says this indication covers both a wake-up detected over the network segment and one detected over a local wake-up pin. The state-machine text further says it is asserted when either a valid **Wake-Up Pulse** is detected on the medium or a valid local wake request is asserted. So this primitive reports **that wake-up has happened**, regardless of whether the source was local or from the wire.

3. Wake forwarding for multiport devices such as switches

- **WakeupForward.indication**
  This means: **"a wake-up forwarding event arrived here."** The spec marks it optional and says it signals that a wake-up forwarding event has been received over wake Input/Output functionality or over the medium-dependent interface. In practical switch language: one side of the device has learned of a wake event that may need to be relayed elsewhere.
- **WakeupForward.request**
  This means: **"forward that wake-up event out through this port."** The spec says this optional primitive signals that a wake-up event has been forwarded to this port as a consequence of a **WakeupForward.indication** on another port or through the wake Input/Output functionality. In other words, **indication = received for forwarding**,

while **request = now propagate it from this port**. The forwarding timing definitions in the spec are built exactly around this pair.

A compact mental model is:

- **LowPowerEntryLocal.request** = sleep **this PHY**
- **WakeupLocal.request** = wake **this PHY**
- **Wakeup.request** = wake **this segment**
- **Wakeup.indication** = **a wake event was detected**
- **WakeupForward.indication** = **a wake event arrived here to be relayed**
- **WakeupForward.request** = **relay that wake event out on this port**

A simple automotive switch example makes the last two clearer.

Suppose a wake event comes into **port A** of a zonal switch from one 10BASE-T1S branch. The switch-side power-management logic sees that as **WakeupForward.indication**.

It then decides that **port B** and **port C** must also wake their attached branches, so it issues **WakeupForward.request** on those ports.

Those ports then propagate the wake-up onward.

The spec explicitly says communication of wake-up events between power-management clients is achieved through the **WakeupForward** primitives.

The most important distinction is this: **Local** primitives control or report the state of the **local PHY**, while **Wakeup.request** and the **WakeupForward** primitives deal with distributing wake-up across a **shared segment** or across **multiple ports** in a multiport device.

That is how it adapts the Technical Committee 10 philosophy to a shared Ethernet segment rather than a two-node link.

The original point-to-point Technical Committee 10 problem is “how do two ends of one link sleep and wake correctly?”

The 10BASE-T1S problem is “how do many nodes on one shared segment sleep and wake without corrupting traffic or confusing each other?”

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13. The 10BASE-T1S Wake-Up Pulse pattern

The public 10BASE-T1S sleep/wake specification gives a much more structured wake signal than the older point-to-point document.

It says the **Wake-Up Pulse** may be sent on either a quiet or a partial segment and is transmitted directly onto the Medium Dependent Interface by the 10BASE-T1S PHY.

The same public text says the Wake-Up Pulse is made of four sections:

- **SUSPEND**
- **Wake-Up Tone**
- **COMMIT**
- **ESD/ESDOK**

The document then gives the actual composition:

- the **SUSPEND** section contains six Differential Manchester Encoding T symbols
- the **Wake-Up Tone** contains twelve periods of a 625 kilohertz tone
- the **COMMIT** section contains 24 to 26 Differential Manchester Encoding J symbols
- total Wake-Up Pulse duration is 32.0 to 32.8 microseconds, with 32.4 microseconds typical

Something fundamental about automotive Ethernet wake-up design: the wake signal must be electrically distinctive enough to be detected robustly at PHY level, yet short enough not to damage bus efficiency.

The public specification also says other nodes on the 10BASE-T1S segment must not start normal transmissions while a Wake-Up Pulse is active.

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14. Quiet segment versus partial segment: why the distinction matters

The public 10BASE-T1S sleep/wake specification defines a **quiet link segment** as one with no physical-medium activity and a **partial link segment** as one with at least one node transmitting on the medium, including Physical Layer Collision Avoidance beacons.

Even if no payload frames are flowing, a segment with beacons is still not “quiet.”

This matters because wake-up timing differs between the two cases.

The public specification says that on a quiet segment, the maximum time to begin transmitting a Wake-Up Pulse after a wake request is bounded by **TWU_Start_quiet**.

On a partial segment, a Physical Layer Collision Avoidance node with good status sends the Wake-Up Pulse during its own transmit opportunity, and the specification gives the bound:

TWU_Start_partial < TWU_Start_quiet(max) + maxPLCACycleTime

That single formula captures a deep engineering truth: once a shared medium is already active, wake-up latency is no longer just a PHY problem.

It is also an arbitration problem. The wake request must respect the transmit-opportunity structure of the segment.

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15. Physical Layer Collision Avoidance pause during wake-up

The 10BASE-T1S sleep/wake specification goes even further and adds a **PLCA Pause** behavior.

Public text says the **paused** variable suspends the generation of transmit opportunities and stops the Reconciliation Sublayer from conveying data to the PHY while it is true.

The same section defines a **resume_timer** with duration 240 bit-times plus or minus 5 bit-times after wake-up reporting ends.

This is one of those details that separates “I know the acronym” from “I understand the system.”

Without a pause concept, normal transmit-opportunity generation could interfere with wake signaling on an active shared segment.

With the pause concept, wake activity temporarily takes control of medium arbitration, and normal Physical Layer Collision Avoidance operation resumes only after the wake condition has been safely handled.

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16. What a bring-up engineer actually has to get right

If one is bringing up a real 10BASE-T1S automotive segment, the practical checklist would be:

1. Need one node configured as coordinator with identifier zero.
2. Need Every other node should have a unique identifier.
3. Need the coordinator’s node-count setting to cover the largest installed population.
4. Need the transmit-opportunity timer to match across all nodes.
5. Need burst mode either disabled or deliberately engineered around your traffic pattern.
6. Need wake behaviour verified both on quiet segments and on partially active segments.
7. If Technical Committee 14 style PMD transceiver interface is being used, need the host controller and transceiver to agree on command timing, receive-pin interpretation, Energy Detect behaviour, reset sequencing, and low-power wake handshakes.

If any one of those is wrong, the failure can look confusingly “Ethernet-like” at first even though the root cause is actually shared-medium orchestration.

Duplicate node identifiers can masquerade as random corruption.

Wrong node count can look like missing access opportunities.

Mismatched transmit-opportunity timing can look like unstable bus behaviour.

Incorrect wake sequencing can make some nodes seem dead while others appear fine.

The public diagnostic bits in the OPEN Alliance register specification exist precisely because these are real field problems, not theoretical edge cases.

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17. Common misconceptions

Misconception 1: 10BASE-T1S is just "10 megabit point-to-point Ethernet on one pair."
No. It can do point-to-point, but its real architectural significance is that it also supports half-duplex multidrop operation over a shared single twisted pair.

Misconception 2: Physical Layer Collision Avoidance replaces Ethernet medium access.
No. Public IEEE material explicitly says it works with Carrier Sense Multiple Access with Collision Detection functions rather than replacing them with a completely separate protocol.

Misconception 3: Physical Layer Collision Avoidance is just Time Division Multiple Access.
No. It gives transmit opportunities, not full fixed packet slots, so idle nodes waste only a short timer window rather than a complete slot.

Misconception 4: "OPEN Alliance TC14 version 1.5 automotive profile" is one public PDF.
Publicly, Technical Committee 14 is a family of specifications. Version 1.5 is clearly associated with the PMD interface and the half-duplex interoperability suite, while other Technical Committee 14 documents have different version numbers.

Misconception 5: TC10 sleep/wake for 10BASE-T1S is identical to classic point-to-point TC10.
No. The 10BASE-T1S sleep/wake document adds a dedicated Power Management client model and a Wake-Up-Pulse-based shared-segment approach tailored to quiet and partial multidrop networks.

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18. Summary

Clause 147 gives the **10BASE-T1S physical layer**.

Clause 148 gives **Physical Layer Collision Avoidance**, which is the practical way to make a half-duplex multidrop Ethernet segment behave predictably enough for automotive use.

OPEN Alliance Technical Committee 14 builds the automotive productization layer on top of that foundation through system integration, host-to-transceiver interface definition, diagnostics, topology discovery, interoperability, and test work.

OPEN Alliance Technical Committee 10 adds the power-state discipline so the network can enter low power and wake in a controlled automotive way. 10BASE-T1S is not merely a low-speed PHY, but a complete low-cost Ethernet edge-networking concept for vehicle